Notes for Chapter 20 of Vakil's FOAG

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Contents

Remark. *Let X be a k-variety. Most applications will consider projective varieties.*

1 Intersecting n line bundles with an n-dimensional variety

Definition 1.1 (20.1.1)**.** Let X be a k-variety, and let $\mathcal{F} \in \mathrm{Coh}(X)$ be a coherent sheaf such that the support of $\mathcal{F}$ has dimension at most n. Let $\mathcal{L}_1, \ldots, \mathcal{L}_n$ be line bundles on X. Then the intersection number is defined as:

$$(\mathcal{L}_1 \ldots \mathcal{L}_n \cdot \mathcal{F}) := \sum_{\{i_1,\ldots,i_m\} \subseteq \{1,\ldots,n\}} (-1)^m \chi(X, \mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F})$$

Definition 1.2 (Notations)**.** Let Y be a closed subscheme of X. We define the intersection number with Y as:

$$(\mathcal{L}_1 \mathcal{L}_2 \ldots \mathcal{L}_n \cdot Y) := (\mathcal{L}_1 \ldots \mathcal{L}_n \cdot \mathcal{O}_Y).$$

If $Y = X$, we simply write $(\mathcal{L}_1 \ldots \mathcal{L}_n) := (\mathcal{L}_1 \ldots \mathcal{L}_n \cdot X)$. Furthermore, we use the shorthand notations $(\mathcal{L}^n \cdot \mathcal{F})$, $(\mathcal{L}^n \cdot Y)$, and $(\mathcal{L}^n)$ when all line bundles are identical.

Proposition 1.3 (20.1.A)**.** If $\mathcal{L}_1 \cong \mathcal{O}_X$, then $(\mathcal{L}_1 \mathcal{L}_2 \ldots \mathcal{L}_n \cdot \mathcal{F}) = 0$.

Proof. For the base case $n = 1$, we have:

$$(\mathcal{L}_1 \cdot \mathcal{F}) = \chi(\mathcal{F}) + (-1)^1 \chi(\mathcal{O}_X \otimes \mathcal{F}) = 0.$$

In general, we can split the sum over subsets depending on whether they contain the index 1. This yields:

$$\begin{aligned}
(\mathcal{L}_1 \ldots \mathcal{L}_n \cdot \mathcal{F}) &= \sum_{1 \notin \{i_1,\ldots,i_m\}} (-1)^m \chi(X, \ldots) + \sum_{1 \in \{i_1,\ldots,i_m\}} (-1)^m \chi(X, \ldots) \\
&= \sum_{1 \notin \{i_1,\ldots,i_m\}} \left((-1)^m + (-1)^{m+1}\right) \chi(X, \ldots) \\
&= 0.
\end{aligned}$$

□

Proposition 1.4 (20.1.B)**.** The intersection number is preserved by field extension of k.

Proof. Let $p : X_K \to X$ be the morphism induced by the field extension K/k. That is, we want to show that:

$$(p^*\mathcal{L}_1 \dots p^*\mathcal{L}_n \cdot p^*\mathcal{F}) = (\mathcal{L}_1\mathcal{L}_2 \dots \mathcal{L}_n \cdot \mathcal{F}).$$

We only need to show that for any coherent sheaf $\mathcal{G} \in \mathrm{Coh}(X)$, we have:

$$\dim_K H^i(X_K, p^*\mathcal{G}) = \dim_k H^i(X, \mathcal{G}).$$

Since X is proper, we can use Čech cohomology to compute both $H^i(X, \mathcal{G})$ and $H^i(X_K, p^*\mathcal{G})$. Consider an affine open cover $\mathcal{U} = \{U_i\}_{i \in I}$ where each U_i is affine. It is then easy to see that:

$$H^i(X_K, p^*\mathcal{G}) \cong H^i(X, \mathcal{G}) \otimes_k K.$$

This implies the dimensions over the respective fields are equal, which concludes the proof. □

Remark. *By this proposition, we can always assume the base field k is infinite when proving properties of the intersection number.*

Proposition 1.5 (20.1.C)**.**

(a) Let X be a projective curve and $\mathcal{L}$ be a line bundle on X. Then $(\mathcal{L} \cdot X) = \deg \mathcal{L}$.
(This holds because $(\mathcal{L} \cdot X) = -\chi(\mathcal{L}^\vee) + \chi(\mathcal{O}_X) = -\deg \mathcal{L}^\vee = \deg \mathcal{L}$ by 18.4.M.)

(b) Let k be an infinite field, $X = \mathbb{P}^N_k$, and Y be a subvariety of X with $\dim Y = n$. We can choose hypersurfaces $H_1, \dots, H_n$ of degrees $d_1, \dots, d_n$ such that $\dim(H_1 \cap H_2 \dots \cap H_n \cap Y) = 0$ (cf. 12.3.D(b)). By Bézout's theorem:

$$(\mathcal{O}_X(H_1) \dots \mathcal{O}_X(H_n) \cdot Y) = \deg(H_1 \cap \dots \cap H_n \cap Y) = d_1 \dots d_n \deg Y.$$

Proof. Let $\iota : Y \hookrightarrow X$ be the inclusion map. Note that because ι is an finite morphism, we have:

$$H^i(Y, \mathcal{L}_{i_1}^\vee|_Y \otimes \dots \otimes \mathcal{L}_{i_m}^\vee|_Y \otimes \mathcal{O}_Y) = H^i(X, \iota_*(\mathcal{L}_{i_1}^\vee|_Y \otimes \dots \otimes \mathcal{L}_{i_m}^\vee|_Y \otimes \mathcal{O}_Y)).$$

By the projection formula, this is isomorphic to:

$$H^i(X, \mathcal{L}_{i_1}^\vee \otimes \dots \otimes \mathcal{L}_{i_m}^\vee \otimes \iota_*\mathcal{O}_Y).$$

Therefore, we can rewrite the intersection number as:

$$(\mathcal{L}_1 \dots \mathcal{L}_n \cdot Y) = \sum_{\{i_1, \dots, i_m\}} (-1)^m \chi(Y, \mathcal{L}_{i_1}^\vee|_Y \otimes \dots \otimes \mathcal{L}_{i_m}^\vee|_Y).$$

Consider the short exact sequence:

$$0 \to \mathcal{L}_n^\vee|_Y \to \mathcal{O}_Y \to \mathcal{O}_{Y\cap H_n} \to 0.$$

Tensoring this sequence with $\left(\bigotimes_{j\neq n}\mathcal{L}_j^\vee\right)\big|_Y$ and utilizing the additive property of the Euler characteristic, we obtain:

$$\begin{aligned}
(\mathcal{L}_1\ldots\mathcal{L}_n\cdot Y) &= \sum_{n\notin\{i_1,\ldots,i_m\}} (-1)^m \left(\chi(\mathcal{L}_{i_1}^\vee|_Y\otimes\cdots\otimes\mathcal{L}_{i_m}^\vee|_Y) - \chi(\mathcal{L}_{i_1}^\vee|_Y\otimes\cdots\otimes\mathcal{L}_{i_m}^\vee|_Y\otimes\mathcal{L}_n^\vee|_Y)\right) \\
&= \sum_{n\notin\{i_1,\ldots,i_m\}} (-1)^m \chi(\mathcal{L}_{i_1}^\vee|_Y\otimes\cdots\otimes\mathcal{L}_{i_m}^\vee|_Y\otimes\mathcal{O}_{Y\cap H_n}) \\
&= (\mathcal{L}_1\ldots\mathcal{L}_{n-1}\cdot Y\cap H_n).
\end{aligned}$$

Applying this reduction repeatedly, we eventually conclude that:

$$(\mathcal{L}_1\ldots\mathcal{L}_n\cdot Y) = (\mathcal{O}\cdot Y\cap H_1\cdots\cap H_n) = \deg(Y\cap H_1\cdots\cap H_n) = d_1\ldots d_n\deg Y.$$

□

We can get one technical consequence from the above proof:

Proposition 1.6 (20.1.D)**.** Let D be an effective Cartier divisor on X, and let $Y\hookrightarrow X$ be a closed subscheme such that $D|_Y = D\cap Y$. Then:

$$(\mathcal{L}_1\ldots\mathcal{L}_{n-1}\cdot\mathcal{O}(D)\cdot Y) = (\mathcal{L}_1\ldots\mathcal{L}_{n-1}\cdot D|_Y).$$

Definition 1.7 (Notation)**.** If $\mathcal{L}_i = \mathcal{O}(D_i)$, we often denote the intersection number simply by using the divisors:

$$(\mathcal{L}_1\ldots\mathcal{L}_n\cdot\mathcal{F}) = (D_1\ldots D_n\cdot\mathcal{F}).$$

Proposition 1.8 (20.1.3)**.** Let X be a projective k-variety and fix a coherent sheaf $\mathcal{F}$. Then the intersection number $(\mathcal{L}_1\ldots\mathcal{L}_n\cdot\mathcal{F})$ is a symmetric multilinear function of the line bundles $\mathcal{L}_1,\ldots,\mathcal{L}_n$.

Remark. *The previous properties held without needing the assumption that* $\dim(\mathit{Supp}\ \mathcal{F})\leq n$*. However, starting with this multilinearity property, that dimension condition becomes essential.*

Remark. *Recall that* $\mathcal{L}_1\otimes\mathcal{L}_2 = \mathcal{O}(D_1)\otimes\mathcal{O}(D_2) = \mathcal{O}(D_1+D_2)$*. Because of this, we can write the tensor product additively as* $\mathcal{L}_1+\mathcal{L}_2$*. The "addition" referred to in the multilinearity of the* *intersection number corresponds to this tensor product of line bundles.*

Proof of 20.1.3. The symmetry of the intersection number is clear directly from the definition. Furthermore, by the base change property (20.1.B) 1.4, we may assume without loss of generality that the base field k is infinite.

We will prove the multilinearity by induction on n. Now we handle the base case for $n = 1$ (**20.1.E**).

We have following **observation**: Let $\mathcal{A}$ be a very ample line bundle on X. This gives a closed immersion $X \hookrightarrow \mathbb{P}$ into a projective space $\mathbb{P}$ such that the pullback of $\mathcal{O}(1)$ is exactly $\mathcal{A}$.

We can choose a hyperplane H of $\mathbb{P}$ such that the dimension of $\operatorname{Supp}(\mathcal{F}) \cap H$ drops by one, or $\dim(\operatorname{Supp}(\mathcal{F}) \cap H) = 0$ (meaning it consists of a finite number of points).

Consider the standard short exact sequence:

$$0 \to \mathcal{A}^\vee \to \mathcal{O}_X \to \mathcal{O}_{X \cap H} \to 0.$$

I **claim** that we can choose a hyperplane H as above such that tensoring this sequence with $\mathcal{F}$ (i.e., applying the functor $- \otimes \mathcal{F}$) remains, in fact, exact.

Supplement to the previous claim: We only need the hyperplane H to avoid the associated primes $\operatorname{Ass}(\mathcal{F})$ of $\mathcal{F}$. This is achievable because the set of associated primes is finite ($|\operatorname{Ass}(\mathcal{F})| < \infty$).

To verify the exactness, we only need to prove that the map $\Gamma(U, \mathcal{F} \otimes \mathcal{A}^\vee) \to \Gamma(U, \mathcal{F})$ is injective for every affine open subset $U \subset X$.

Locally on an affine open $U = \operatorname{Spec}(R)$, let $\mathcal{F}$ be associated to an R-module M (so $\mathcal{F}|_U = \widetilde{M}$), and let $\mathcal{A}^\vee$ be represented by the ideal generated by a regular element $r \in R$ (so $\mathcal{A}^\vee|_U = \widetilde{rR}$). We need to show that the natural map

$$M \otimes_R (rR) \longrightarrow M, \quad m \otimes rt \longmapsto rtm$$

is injective.

Suppose $\sum(m_i \otimes rt_i) \mapsto 0$. This means $r \cdot (\sum m_i t_i) = 0$. Note that we can rewrite the element in the tensor product as $\sum(m_i \otimes rt_i) = (\sum m_i t_i) \otimes r$. To prove that $\sum m_i t_i = 0$, it suffices to show that the multiplication map $M \xrightarrow{\cdot r} M$ is injective.

Recall from commutative algebra that the set of zero divisors of a module M is the union of its associated primes, $\bigcup_{\mathfrak{p} \in \operatorname{Ass}(M)} \mathfrak{p}$. If multiplication by r is not injective, then r must be a zero divisor. Consequently, there exists an associated prime $\mathfrak{p} \in \operatorname{Ass}(M)$ such that $r \in \mathfrak{p}$.

Geometrically, this implies that the point $[\mathfrak{p}]$ lies in $V(r) \cap \operatorname{Supp}(\widetilde{M})$, which corresponds to the intersection $H \cap \operatorname{Supp}(\mathcal{F})$. However, this is a contradiction, since we explicitly chose the hyperplane H to avoid the associated primes $\mathrm{Ass}(\mathcal{F})$.

Therefore, the multiplication is injective, which implies that the sequence

$$0 \longrightarrow \mathcal{F} \otimes \mathcal{A}^\vee \longrightarrow \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{O}_{X\cap H} \longrightarrow 0$$

is indeed exact. Taking the Euler characteristic yields:

$$\chi(\mathcal{F}) - \chi(\mathcal{A}^\vee \otimes \mathcal{F}) = \chi(\mathcal{F}|_{X\cap H}).$$

Because the support of $\mathcal{F}|_{X\cap H}$ has dimension 0, tensoring it with any line bundle does not change its Euler characteristic, nor does it alter its support. Hence, for any arbitrary line bundle $\mathcal{L}$, we have:

$$\begin{aligned} \chi(\mathcal{L}^\vee \otimes \mathcal{F}) - \chi(\mathcal{A}^\vee \otimes \mathcal{L}^\vee \otimes \mathcal{F}) &= \chi((\mathcal{L}^\vee \otimes \mathcal{F})|_{X\cap H}) \\ &= \chi(\mathcal{F}|_{X\cap H}) \\ &= \chi(\mathcal{F}) - \chi(\mathcal{A}^\vee \otimes \mathcal{F}). \end{aligned}$$

This implies that the proposition holds when one of the line bundles is very ample, i.e.:

$$(\mathcal{L}_1 \cdot \mathcal{F}) + (\mathcal{L}_2 \cdot \mathcal{F}) = ((\mathcal{L}_1 \otimes \mathcal{L}_2) \cdot \mathcal{F}) \quad \text{holds when one of } \mathcal{L}_i \text{ is very ample.}$$

For arbitrary line bundles $\mathcal{L}_1$ and $\mathcal{L}_2$, there exist very ample line bundles $\mathcal{A}_i$ and $\mathcal{B}_i$ such that $\mathcal{L}_i \cong \mathcal{A}_i \otimes \mathcal{B}_i^\vee$.

Since $(\mathcal{O}_X \cdot \mathcal{F}) = 0$, we have $(\mathcal{B}_i \cdot \mathcal{F}) + (\mathcal{B}_i^\vee \cdot \mathcal{F}) = 0$, which implies $(\mathcal{B}_i^\vee \cdot \mathcal{F}) = -(\mathcal{B}_i \cdot \mathcal{F})$. Therefore, we can write:

$$(\mathcal{L}_i \cdot \mathcal{F}) = (\mathcal{A}_i \cdot \mathcal{F}) + (\mathcal{B}_i^\vee \cdot \mathcal{F}) = (\mathcal{A}_i \cdot \mathcal{F}) - (\mathcal{B}_i \cdot \mathcal{F}).$$

By substituting these expressions and expanding, the multilinearity holds for the base case $n = 1$.

Now, assume the proposition is correct for integers strictly less than n. We wish to show it holds for a coherent sheaf $\mathcal{F} \in \mathrm{Coh}(X)$ with $\dim(\mathrm{Supp}\ \mathcal{F}) \le n$. Specifically, we want to prove:

$$(\mathcal{L}_1\mathcal{L}_2 \dots \mathcal{L}_n \cdot \mathcal{F}) + (\mathcal{L}_1'\mathcal{L}_2 \dots \mathcal{L}_n \cdot \mathcal{F}) = ((\mathcal{L}_1 \otimes \mathcal{L}_1')\mathcal{L}_2 \dots \mathcal{L}_n \cdot \mathcal{F}) \tag{1}$$

Note that we can expand the intersection number of $n+1$ line bundles by partitioning the sum based on the inclusion of the first two indices:

$$\begin{aligned}(\mathcal{L}_1\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}) &= \sum_{\text{excluding } 1'} + \sum_{\text{excluding } 1} + \sum_{\text{including both 1 and } 1'} - \sum_{\text{excluding both 1 and } 1'} \\ &= (\mathcal{L}_1\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}) + (\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}) - ((\mathcal{L}_1\otimes\mathcal{L}_1')\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}).\end{aligned}$$

This establishes the relation (**20.1.F**).

Remark. *Because* $\dim(\mathit{Supp}\ \mathcal{F}) \leq n < n+1$, *the term on the left-hand side is well-defined.*

Next, suppose $\mathcal{L}_n = \mathcal{O}(D)$, where D is an effective Cartier divisor such that $D \cap \mathrm{Ass}(\mathcal{F}) = \emptyset$. From the analysis above, tensoring the short exact sequence

$$0 \longrightarrow \mathcal{O}(-D) = \mathcal{L}_n^\vee \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0$$

with $\mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F}$ preserves exactness. By a similar conclusion as in Proposition 1.6 (20.1.D), we obtain:

$$(\mathcal{L}_1\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}) = (\mathcal{L}_1|_D\ldots\mathcal{L}_{n-1}|_D\cdot\mathcal{F}|_D).$$

Similar reductions apply for $(\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F})$ and $((\mathcal{L}_1\otimes\mathcal{L}_1')\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F})$. Since the dimension of the support strictly decreases when restricting to D, we can then apply the induction hypothesis to these reduced terms:

$$\begin{aligned}&(\mathcal{L}_1\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}) + (\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}) - ((\mathcal{L}_1\otimes\mathcal{L}_1')\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}) \\ &= (\mathcal{L}_1\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F}) \\ &= (\mathcal{L}_1\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_{n-1}\cdot\mathcal{F}|_D) \\ &= (\mathcal{L}_1\mathcal{L}_2\ldots\mathcal{L}_{n-1}\cdot\mathcal{F}|_D) + (\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_{n-1}\cdot\mathcal{F}|_D) - ((\mathcal{L}_1\otimes\mathcal{L}_1')\mathcal{L}_2\ldots\mathcal{L}_{n-1}\cdot\mathcal{F}|_D) \\ &= 0 \quad \text{by the induction hypothesis.}\end{aligned}$$

We can successfully perform the induction step whenever some $\mathcal{L}_i$ with $i \neq 1$ is of the form $\mathcal{O}(D)$ with the property that $D \cap \mathrm{Ass}(\mathcal{F}) = \emptyset$ (we've done **20.1.G**).

We hope that if $\mathcal{L}_1 = \mathcal{O}(D)$ with $D \cap \mathrm{Ass}(\mathcal{F}) = \emptyset$, we can still apply the induction hypothesis and get the multilinearity. Using the identity established in our previous note, when

$\mathcal{L}_1$ takes this form, we can evaluate the multilinearity expansion:

$$\begin{aligned}
&(\mathcal{L}_1\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F})+(\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F})-((\mathcal{L}_1\otimes\mathcal{L}_1')\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F})\\
&=(\mathcal{L}_1\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F})\\
&=(\mathcal{L}_1'\mathcal{L}_2\mathcal{L}_3\ldots\mathcal{L}_n\mathcal{L}_1\cdot\mathcal{F})\quad\text{(by symmetry)}\\
&=(\mathcal{L}_1'\mathcal{L}_3\ldots\mathcal{L}_n\mathcal{L}_1\cdot\mathcal{F})+(\mathcal{L}_2\mathcal{L}_3\ldots\mathcal{L}_n\mathcal{L}_1\cdot\mathcal{F})-((\mathcal{L}_1'\otimes\mathcal{L}_2)\mathcal{L}_3\ldots\mathcal{L}_n\mathcal{L}_1\cdot\mathcal{F})\\
&=0.
\end{aligned}$$

Hence by induction we can conclude that (1) holds for the case where $\mathcal{L}_1$ is of the form $\mathcal{O}(D)$ with $D\cap\mathrm{Ass}(\mathcal{F})=\emptyset$. For general case, write $\mathcal{L}_1=\mathcal{A}_1\otimes\mathcal{B}_1^\vee$ and $\mathcal{L}_1'=\mathcal{A}_1'\otimes\mathcal{B}_1'^\vee$ "as the difference of two very amples". Then the proof can be done by the method similar to the case $n=1$. □

Corollary 1.9 (20.1.H)**.** Let X be a projective variety and let $\mathcal{F}$ be a coherent sheaf such that $\dim(\mathrm{Supp}\ \mathcal{F})\leq n<n+1$. For any line bundles $\mathcal{L}_1,\mathcal{L}_1',\mathcal{L}_2,\ldots,\mathcal{L}_n$, we have:

$$(\mathcal{L}_1\mathcal{L}_1'\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F})=0.$$

Proof. This follows immediately from the relation established in (20.1.F) and the multilinearity from Proposition 20.1.3. □

Proposition 1.10 (20.1.6)**. Recall:** Two line bundles are numerically equivalent, denoted as $\mathcal{L}_1\equiv_{\mathrm{num}}\mathcal{L}_2$, if for every curve C, we have:

$$\deg_C(\mathcal{L}_1|_C)=\deg_C(\mathcal{L}_2|_C).$$

As derived earlier in Proposition 20.1.C, this is equivalent to saying $(\mathcal{L}_1\cdot C)=(\mathcal{L}_2\cdot C)$. The proposition states that the intersection number $(\mathcal{L}_1\mathcal{L}_2\ldots\mathcal{L}_n\cdot\mathcal{F})$ only depends on the numerical classes of the line bundles $\mathcal{L}_i$.

Remark. *We omit the proof here and you can check [FOAG] for details. The conclusion is so important and natural that it is not worth spending too much time on the technical details of the proof.*

Theorem 1.11 (**Asymptotic Riemann-Roch**, 20.1.I)**.** Let X be a projective variety, and let $\mathcal{F}\in\mathrm{Coh}(X)$ be a coherent sheaf such that $\dim(\mathrm{Supp}\ \mathcal{F})\leq n$. Let $\mathcal{L}$ be a line bundle on X. Then the function defined by

$$q(m):=\chi(X,\mathcal{L}^{\otimes m}\otimes\mathcal{F})$$

is a polynomial in m of degree at most n, and the coefficient of the m^n term is given by $\frac{1}{n!}(\mathcal{L}^n \cdot \mathcal{F})$.

Proof. By Corollary 20.1.H, since we are intersecting $n+1$ line bundles and the dimension of the support of $\mathcal{L}^{\otimes m} \otimes \mathcal{F}$ is at most n, the intersection number vanishes:

$$(\underbrace{\mathcal{L}\mathcal{L}\ldots\mathcal{L}}_{n+1} \cdot (\mathcal{L}^{\otimes m} \otimes \mathcal{F})) = 0.$$

By the definition of the intersection number and the multilinearity, we obtain:

$$\sum_{j=0}^{n+1} (-1)^{n+1-j} \binom{n+1}{j} \chi(X, \mathcal{L}^{\otimes(m+j)} \otimes \mathcal{F}) = \sum_{j=0}^{n+1} (-1)^{n+1-j} \binom{n+1}{j} q(m+j) = 0$$

for all integers $m \in \mathbb{Z}$. In fact, this alternating sum represents the $(n+1)$-th finite difference of the function $q(m)$, commonly denoted as $\Delta^{n+1} q(m)$.

Since $\Delta^{n+1} q(m) = 0$ for all m, we conclude that $q(m)$ must be a polynomial in m of degree at most n.

Let us write this polynomial explicitly as $q(m) = \sum_{i=0}^{n} c_i m^i$. A standard property of the finite difference operator states that the n-th finite difference of a polynomial of degree n is constant and is given by $\Delta^n q(m) = n! \cdot c_n$.

Now, applying the definition of the intersection number again for n copies of the line bundle $\mathcal{L}$, we find that:

$$(\underbrace{\mathcal{L}\mathcal{L}\ldots\mathcal{L}}_{n} \cdot (\mathcal{L}^{\otimes m} \otimes \mathcal{F})) = \Delta^n q(m).$$

Evaluating it at $m = 0$ gives $(\mathcal{L}^n \cdot \mathcal{F})$. Equating this with the n-th finite difference yields:

$$(\mathcal{L}^n \cdot \mathcal{F}) = n! \cdot c_n.$$

Therefore, the leading coefficient of the polynomial is precisely $c_n = \frac{1}{n!}(\mathcal{L}^n \cdot \mathcal{F})$, which completes the proof. □

Corollary 1.12. When X is a projective curve, we have:

$$\chi(X, \mathcal{F} \otimes \mathcal{L}^{\otimes m}) = (\mathcal{L} \cdot \mathcal{F}) \cdot m + \chi(X, \mathcal{F}) = \deg(\mathcal{L}) \cdot \operatorname{rank} \mathcal{F}^{1} \cdot m + \chi(X, \mathcal{F}).$$

[1] Similarly we can assume $\mathcal{L} = \mathcal{O}(D)$ is very ample for some divisor D. By X is a curve, D must be of the form $\sum a_i p_i$ for some points p_i and integers $a_i > 0$. There is a natural fact that we can choose a D such that $D \cap \operatorname{Ass}(\mathcal{F}) = \emptyset$ and $\mathcal{F}$ is locally free at every p_i. Then $(\mathcal{L} \cdot \mathcal{F}) = \sum_i a_i \cdot h^0(\mathcal{F}|_{p_i}) = \deg \mathcal{L} \cdot \operatorname{rank} \mathcal{F}$.

We should **acknowledge** the following useful projection formula:

Theorem 1.13 (Projection Formula)**.** Let $\pi : X_1 \longrightarrow X_2$ be a morphism between integral projective varieties of dimension n (note that such a morphism is automatically projective). Let $\mathcal{L}_1, \dots, \mathcal{L}_n$ be line bundles on X_2. For any coherent sheaf $\mathcal{F}$ on X_1, the following intersection formula holds:

$$(\pi^*\mathcal{L}_1 \dots \pi^*\mathcal{L}_n \cdot \mathcal{F}) = (\mathcal{L}_1 \dots \mathcal{L}_n \cdot \pi_*\mathcal{F}).$$

Proof Sketch.

$$\begin{aligned}
(\pi^*\mathcal{L}_1 \dots \pi^*\mathcal{L}_n \cdot \mathcal{F}) &= \sum_{\{i_1,\dots,i_m\}} (-1)^m \chi(X_1, \pi^*\mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \pi^*\mathcal{L}_{i_m}^\vee \otimes \mathcal{F}) \\
&= \sum_{\{i_1,\dots,i_m\}} (-1)^m \left(\sum_{j=0}^{\infty} \chi\left(X_2, R^j\pi_*(\pi^*\mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \pi^*\mathcal{L}_{i_m}^\vee \otimes \mathcal{F})\right) \right) \\
&= \sum_{\{i_1,\dots,i_m\}} (-1)^m \left(\sum_{j=0}^{\infty} \chi\left(X_2, \mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \mathcal{L}_{i_m}^\vee \otimes R^j\pi_*\mathcal{F}\right) \right) \\
&= \sum_{j=0}^{\infty} (\mathcal{L}_1 \dots \mathcal{L}_n \cdot R^j\pi_*\mathcal{F}) \\
&= (\mathcal{L}_1 \dots \mathcal{L}_n \cdot \pi_*\mathcal{F}).
\end{aligned}$$

In the second equality we use one equation:

$$\chi(X_1, \mathcal{F}) = \sum_{j=0}^{\infty} \chi(X_2, R^j\pi_*\mathcal{F}) \quad \forall \mathcal{F} \in \mathrm{Coh}(X_1),$$

and the last equality follows from $\dim \mathrm{supp}(R^j\pi_*\mathcal{F}) < \dim \mathrm{supp}(\mathcal{F})$ for all $j > 0$. $\square$

Corollary 1.14 (20.1.J)**.** $\pi : X_1 \longrightarrow X_2$, $\mathcal{L}_1, \dots, \mathcal{L}_n$ as above. Then we have:

$$(\pi^*\mathcal{L}_1 \dots \pi^*\mathcal{L}_n) = \deg(X_1/X_2) \cdot (\mathcal{L}_1 \dots \mathcal{L}_n).$$

Proof Sketch. From the general projection formula, we have:

$$(\mathcal{L}_1\mathcal{L}_2 \dots \mathcal{L}_n \cdot \pi_*\mathcal{O}_{X_1}) = (\pi^*\mathcal{L}_1 \dots \pi^*\mathcal{L}_n \cdot \mathcal{O}_{X_1}).$$

Thus, we need to show that:

$$(\mathcal{L}_1\mathcal{L}_2 \dots \mathcal{L}_n \cdot \pi_*\mathcal{O}_{X_1}) = \deg(X_1/X_2) \cdot (\mathcal{L}_1 \dots \mathcal{L}_n \cdot \mathcal{O}_{X_2}).$$

Case 1: Suppose π is not dominant.
The right-hand side is trivially 0 because $\deg(X_1/X_2) = 0$. For the left-hand side, since π is a projective morphism, its image is a closed subset, which means $\mathrm{Supp}(\pi_*\mathcal{O}_{X_1}) = \overline{\pi(X_1)} = \pi(X_1)$. Because π is not dominant, we have $\pi(X_1) \subsetneq X_2$, which implies that $\dim(\mathrm{Supp}(\pi_*\mathcal{O}_{X_1})) < n$. Therefore, intersecting n line bundles with it yields 0, so the left-hand side is also 0.

Case 2: Suppose π is dominant.
Since both the left-hand side and the right-hand side are multilinear, it suffices to prove the case when the line bundles $\mathcal{L}_1, \dots, \mathcal{L}_n$ are all very ample.

(i) We can choose an open subset $U' \subseteq X_2$ such that the restriction $\pi|_{\pi^{-1}(U')}$ is finite (cf. 10.3.G). Then, we can choose a smaller open subset $U \subseteq U'$ such that $\pi_*\mathcal{O}_{X_1}|_U$ is locally free of rank $\deg(X_1/X_2)$ (cf. 14.3.E). (Alternatively, by Chapter 24, there exists a dense open subset $U \subseteq X_2$ such that $\pi|_{\pi^{-1}(U)}$ is finite and flat).

(ii) From the choice of U, we clearly have $U \subseteq \mathrm{Supp}(\pi_*\mathcal{O}_{X_1})$. Let $Z = X_2 \setminus U$. By 12.3.D, we can choose divisors $D_1 \sim \mathcal{L}_1, \dots, D_n \sim \mathcal{L}_n$ such that:

- $\dim(D_1 \cap \dots \cap D_n) = 0$,
- $(D_1 \cap \dots \cap D_n) \cap Z = \emptyset$,
- The intersections avoid the associated primes $\mathrm{Ass}(\pi_*\mathcal{O}_{X_1})$.

By the previous theorems, we can compute the intersection number by restricting to these divisors:

$$(\mathcal{L}_1 \dots \mathcal{L}_n \cdot \pi_*\mathcal{O}_{X_1}) = \chi(\pi_*\mathcal{O}_{X_1}|_{D_1 \cap \dots \cap D_n}).$$

Note that the intersection $D_1 \cap \dots \cap D_n$ consists of finitely many discrete points. Because $(D_1 \cap \dots \cap D_n) \cap Z = \emptyset$, all these points lie entirely in U. Over the open set U, $\pi_*\mathcal{O}_{X_1}$ is locally free of rank $\deg(X_1/X_2)$. Therefore, taking the Euler characteristic simply multiplies it by this rank:

$$\begin{aligned}
\chi(\pi_*\mathcal{O}_{X_1}|_{D_1 \cap \dots \cap D_n}) &= \chi(\mathcal{O}_{X_2}|_{D_1 \cap \dots \cap D_n}) \cdot \mathrm{rank}(\pi_*\mathcal{O}_{X_1}|_U) \\
&= \chi(\mathcal{O}_{X_2}|_{D_1 \cap \dots \cap D_n}) \cdot \deg(X_1/X_2) \\
&= (\mathcal{L}_1 \dots \mathcal{L}_n \cdot \mathcal{O}_{X_2}) \cdot \deg(X_1/X_2).
\end{aligned}$$

This completes the proof. $\square$

Proposition 1.15 (20.1.K)**.** Let X be a projective variety, and let Y be a subvariety of X of dimension n. If $\mathcal{L}_1, \ldots, \mathcal{L}_n$ are ample line bundles on X, then we have:

$$(\mathcal{L}_1 \ldots \mathcal{L}_n \cdot Y) > 0.$$

Proof. We proceed by induction on n.

For the base case $n = 1$, Y is a projective curve. We know that a line bundle on a curve is ample if and only if its degree is strictly positive. Therefore, the intersection number is simply the degree, which gives:

$$(\mathcal{L}_1 \cdot Y) = \deg_Y(\mathcal{L}_1|_Y) > 0.$$

Now, assume the proposition holds for dimension $n-1$. Since $\mathcal{L}_n$ is ample, there exists a sufficiently large integer $m > 0$ such that the tensor power $\mathcal{L}_n^{\otimes m}$ becomes very ample.

By the multilinearity of the intersection number, we can write:

$$m(\mathcal{L}_1\mathcal{L}_2 \ldots \mathcal{L}_n \cdot Y) = (\mathcal{L}_1\mathcal{L}_2 \ldots \mathcal{L}_{n-1}\mathcal{L}_n^{\otimes m} \cdot Y).$$

Because $\mathcal{L}_n^{\otimes m}$ is very ample, we can choose an effective Cartier divisor D such that $\mathcal{L}_n^{\otimes m} \cong \mathcal{O}(D)$ and D intersects Y properly. Then the intersection number becomes:

$$(\mathcal{L}_1\mathcal{L}_2 \ldots \mathcal{L}_{n-1}\mathcal{O}(D) \cdot Y) = (\mathcal{L}_1\mathcal{L}_2 \ldots \mathcal{L}_{n-1} \cdot D \cap Y).$$

Notice that $D \cap Y$ is a closed subscheme of Y of dimension $n-1$. The inclusion map $D \cap Y \hookrightarrow Y$ is a finite morphism (specifically, a closed immersion). Since the pullback via a finite morphism preserves the ampleness of line bundles, the restrictions of $\mathcal{L}_1, \ldots, \mathcal{L}_{n-1}$ to $D \cap Y$ are still ample line bundles.

By our induction hypothesis on the $(n-1)$-dimensional variety $D \cap Y$, we have:

$$(\mathcal{L}_1\mathcal{L}_2 \ldots \mathcal{L}_{n-1} \cdot D \cap Y) > 0.$$

Since $m > 0$, this immediately implies that $(\mathcal{L}_1 \ldots \mathcal{L}_n \cdot Y) > 0$, which completes the proof. □

2 Intersection theory on a surface

Remark. *Let X be a regular projective surface. (In fact, it is perhaps sufficient to only assume that X is a Noetherian factorial scheme).*

Definition 2.1 (Notations)**.** Let C and D be effective divisors on X. We define the intersection number of C and D, denoted as $(C \cdot D)$, by the following equality:

$$\deg_C(\mathcal{O}(D)|_C) = -(\mathcal{O}(-D) \cdot C) = (\mathcal{O}(D) \cdot C) = (\mathcal{O}(D) \cdot \mathcal{O}(C) \cdot X) = \deg_D(\mathcal{O}(C)|_D).$$

Furthermore, if we assume that $C \cap \operatorname{Ass}(\mathcal{O}_D) = \emptyset$, then tensoring the short exact sequence

$$0 \longrightarrow \mathcal{O}(-C) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_C \longrightarrow 0$$

with $\mathcal{O}_D$ preserves exactness. This implies:

$$\deg_D(\mathcal{O}(C)|_D) = -\deg_D(\mathcal{O}(-C)|_D) = \chi(\mathcal{O}_{C\cap D}) = h^0(C \cap D, \mathcal{O}_{C\cap D}).$$

Remark (20.2.1)**.** *When X is a Cohen-Macaulay surface (for example, a regular surface), there are automatically no embedded associated points. In this case, the condition $C \cap \operatorname{Ass}(\mathcal{O}_D) = \emptyset$ can be safely replaced by the simpler geometric condition that C and D have no common irreducible components. (Because* Ass $\setminus$ Embedded *corresponds to those minimal associated primes.) Especially, if C and D are both irreducible and $C \neq D$ set-theoretically, then $(C \cdot D) = h^0(C \cap D) \geq 0$.*

Remark (20.2.3)**.** *Recall the definition of the normal bundle $\mathcal{N}_{C/X} := \mathcal{O}_X(C)|_C$. Then its degree is given by the self-intersection number:*

$$\deg \mathcal{N}_{C/X} = (C \cdot C).$$

Remark (20.2.4)**.** *As seen by the definition, the intersection number $(C \cdot D)$ can be expressed in several different ways, each with its own characteristics:*

1. *$(\mathcal{O}(C) \cdot \mathcal{O}(D) \cdot X)$: This expression is very useful for formal manipulations, but it is geometrically somewhat abstract.*

2. *$\deg_D(\mathcal{O}(C)|_D)$: This expression is not manifestly symmetric.*

3. *$h^0(C \cap D, \mathcal{O}_{C\cap D})$: This expression is geometrically intuitive.*

Example 2.2. Let $X = \mathbb{A}_k^2$. Consider the curves $C = V(y - x^2)$ and $D = V(y)$. The intersection scheme is given by:

$$C \cap D \cong \operatorname{Spec} k[x,y]/(y - x^2, y) \cong \operatorname{Spec} k[x]/(x^2).$$

Thus, we have $h^0 = 2$. This correctly reflects the geometric intuition of intersection multiplicity, as the line D is tangent to the parabola C.

By definition, this intersection pairing induces a bilinear form on the Picard group:

$$\begin{aligned} \operatorname{Pic}(X) \times \operatorname{Pic}(X) &\longrightarrow \mathbb{Z} \\ (\mathcal{O}(C), \mathcal{O}(D)) &\longmapsto (C \cdot D). \end{aligned}$$

Theorem 2.3 (Application 1: Recovering Bézout's Theorem)**.** Let C and D be curves in $\mathbb{P}^2$ that share no common components. Then the degree of their intersection is given by:

$$\deg(C \cap D) = \deg C \cdot \deg D.$$

Proof. Let $c = \deg C$ and $d = \deg D$. Since C and D have no common components, the intersection is a proper intersection of dimension zero. The left-hand side is the number of intersection points counted with multiplicity, which equals the Euler characteristic of the intersection scheme:

$$\begin{aligned} \text{LHS} &= h^0(C \cap D, \mathcal{O}_{C \cap D}) \\ &= (\mathcal{O}(C) \cdot \mathcal{O}(D) \cdot \mathcal{O}_{\mathbb{P}^2}) \\ &= (\mathcal{O}(c) \cdot \mathcal{O}(d) \cdot \mathcal{O}_{\mathbb{P}^2}). \end{aligned}$$

Using the general formula for the intersection of two line bundles, $(\mathcal{L}_1 \cdot \mathcal{L}_2) = \chi(\mathcal{L}_1 \otimes \mathcal{L}_2) - \chi(\mathcal{L}_1) - \chi(\mathcal{L}_2) + \chi(\mathcal{O}_X)$, and the fact that $\chi(\mathcal{O}_{\mathbb{P}^2}(n)) = \frac{(n+1)(n+2)}{2}$, we can compute:

$$\begin{aligned} (\mathcal{O}(c) \cdot \mathcal{O}(d)) &= \chi(\mathcal{O}(c+d)) - \chi(\mathcal{O}(c)) - \chi(\mathcal{O}(d)) + \chi(\mathcal{O}_{\mathbb{P}^2}) \\ &= \frac{(c+d+1)(c+d+2)}{2} - \frac{(c+1)(c+2)}{2} - \frac{(d+1)(d+2)}{2} + 1 \\ &= cd. \end{aligned}$$

This completes the proof. □

Proposition 2.4 (Application 2: 20.2.B)**.** Assuming Serre's duality for a smooth projective surface X, the following holds:

1. **Adjunction Formula:** For any curve $C \subset X$,

$$C \cdot (K_X + C) = 2p_a(C) - 2.$$

2. **Riemann-Roch for Surfaces:** For any divisor D on X,

$$\chi(X, \mathcal{O}_X(D)) = \frac{D \cdot (D - K_X)}{2} + \chi(X, \mathcal{O}_X).$$

Proof. **Proof of (1):** Consider the standard short exact sequence $0 \to \mathcal{O}_X(-C) \to \mathcal{O}_X \to \mathcal{O}_C \to 0$. We evaluate the intersection number $C \cdot (-K_X - C)$ using the definition:

$$C \cdot (-K_X - C) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(K_X + C)) - \chi(\mathcal{O}_X(-C)) + \chi(\mathcal{O}_X(K_X)).$$

By Serre duality, we know that $\chi(\mathcal{O}_X(K_X + C)) = \chi(\mathcal{O}_X(-C))$ and $\chi(\mathcal{O}_X(K_X)) = \chi(\mathcal{O}_X)$. Substituting these into the equation yields:

$$\begin{aligned} C \cdot (-K_X - C) &= 2\big(\chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-C))\big) \\ &= 2\chi(\mathcal{O}_C) \quad \text{(from the short exact sequence)} \\ &= 2(1 - p_a(C)). \end{aligned}$$

Multiplying both sides by -1 gives $C \cdot (K_X + C) = 2p_a(C) - 2$.

Proof of (2): First, we compute $-(D \cdot D)$ by treating it as $(D \cdot -D)$:

$$-(D \cdot D) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D)) - \chi(\mathcal{O}_X(D)) + \chi(\mathcal{O}_X) = 2\chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(D)) - \chi(\mathcal{O}_X(-D)).$$

Next, we compute $D \cdot K_X$ by writing it as $(-D) \cdot (-K_X)$:

$$D \cdot K_X = (-D) \cdot (-K_X) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(D)) - \chi(\mathcal{O}_X(K_X)) + \chi(\mathcal{O}_X(K_X + D)).$$

Applying Serre duality again, $\chi(\mathcal{O}_X(K_X)) = \chi(\mathcal{O}_X)$ and $\chi(\mathcal{O}_X(K_X+D)) = \chi(\mathcal{O}_X(-D))$. Thus, the equation simplifies to:

$$D \cdot K_X = \chi(\mathcal{O}_X(-D)) - \chi(\mathcal{O}_X(D)).$$

Adding the expression for $-(D \cdot D)$ and $D \cdot K_X$ together, we obtain:

$$\begin{aligned} D \cdot (K_X - D) &= \big(\chi(\mathcal{O}_X(-D)) - \chi(\mathcal{O}_X(D))\big) + \big(2\chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(D)) - \chi(\mathcal{O}_X(-D))\big) \\ &= 2\chi(\mathcal{O}_X) - 2\chi(\mathcal{O}_X(D)). \end{aligned}$$

Rearranging this equation directly recovers the Riemann-Roch formula for surfaces:

$$\chi(X, \mathcal{O}_X(D)) = \frac{D \cdot (D - K_X)}{2} + \chi(X, \mathcal{O}_X).$$

□

Theorem 2.5 (20.2.C)**.** We know that $\mathrm{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) = \mathbb{Z} \oplus \mathbb{Z}$. Let $l = \mathbb{P}^1 \times \{0\}$ and $m = \{0\} \times \mathbb{P}^1$. The line bundles $\mathcal{O}(l)$ and $\mathcal{O}(m)$ are the generators. Then we have the following intersection numbers:

$$l \cdot l = m \cdot m = 0, \quad l \cdot m = 1.$$

***Proof*.** The intersection $l \cap m$ is exactly one point (which corresponds to Spec k), so $l \cdot m = 1$.

Let $l' = \mathbb{P}^1 \times \{\infty\}$. We have $l \cap l' = \emptyset$, which implies $l \cdot l' = 0$. Since l is linearly equivalent to l', it follows that $l \cdot l = l \cdot l' = 0$. By a symmetric argument, we also obtain $m \cdot m = 0$. □

Application: Consider the diagonal $\Delta \cong \mathbb{P}^1 \hookrightarrow \mathbb{P}^1 \times \mathbb{P}^1$. What are the coefficients a and b such that the class $[\Delta] = a[l] + b[m]$?

By computing the intersection numbers, we have $\Delta \cdot l = 1$ and $\Delta \cdot m = 1$. Expanding the intersection yields $a = 1$ and $b = 1$.

Theorem 2.6 (20.2.D)**.** Let $X = \mathrm{Bl}_p\mathbb{P}^2$ be the blow-up of $\mathbb{P}^2$ at a point p. Let l be a line in $\mathbb{P}^2$ not passing through p (identified with its strict transform), and let e be the exceptional divisor. Then $\mathrm{Pic}(X) = \mathbb{Z}l \oplus \mathbb{Z}e$, and their intersection numbers are:

$$l \cdot l = 1, \quad e \cdot e = -1, \quad l \cdot e = 0.$$

***Proof Sketch*.** This proof relies on a more geometric approach. Consider the blow-up morphism $\pi : \mathrm{Bl}_p\mathbb{P}^2 \to \mathbb{P}^2$. Since the line $l \subset \mathbb{P}^2$ does not pass through p, its strict transform naturally avoids the exceptional divisor.

We can find another line l' in $\mathbb{P}^2$ that passes through p. Since any two lines in $\mathbb{P}^2$ are linearly equivalent, we have $\mathcal{O}_X(l) \cong \mathcal{O}_X(l')$. The pullback of l' is given by $\pi^* l' = e + m$, where m is the strict transform of l'.

First, we determine the self-intersection of m. We can find another line l'' passing through p, whose pullback is $\pi^*l'' = e + m'$. Since $\pi^*l' \sim \pi^*l''$, it follows that $m \sim m'$. We can choose l' and l'' such that their strict transforms do not intersect, meaning $m \cap m' = \emptyset$. Therefore, $m \cdot m = 0$.

Now we compute the remaining intersections:

- $e \cdot m = 1$ (because the strict transform m intersects the exceptional divisor e at exactly one point).
- $l \cdot e = 0$ (because the line l does not pass through p, so its strict transform is disjoint from e).
- $l \cdot m = 1$ (because the intersection $l \cap l'$ is exactly one point in $\mathbb{P}^2$ that is distinct from p).

Using these relations, we can deduce the self-intersection of l:

$$l \cdot l = l \cdot \pi^*l' = l \cdot (e + m) = l \cdot e + l \cdot m = 0 + 1 = 1.$$

For the exceptional divisor, since $\pi^*l' \sim l$, we substitute $e = \pi^*l' - m$:

$$e \cdot e = (\pi^*l' - m) \cdot e = l \cdot e - m \cdot e = 0 - 1 = -1.$$

By the excision exact sequence, we have:

$$\mathbb{Z}e \longrightarrow \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(X \setminus e) \longrightarrow 0.$$

Notice that $X \setminus e \cong \mathbb{P}^2 \setminus \{p\}$. Since the point p has codimension strictly greater than 1 in $\mathbb{P}^2$, the Picard group remains unaffected by its removal. Thus, we have:

$$\mathrm{Pic}(X \setminus e) = \mathrm{Pic}(\mathbb{P}^2 \setminus \{p\}) \cong \mathrm{Pic}(\mathbb{P}^2) \cong \mathbb{Z}.$$

Therefore, $\mathrm{Pic}(X)$ can be generated by the classes of l and e, and their intersection numbers are as stated.

Finally, from the intersection numbers, we can verify that l and e are linearly independent in $\mathrm{Pic}(X)$, confirming that $\mathrm{Pic}(X) = \mathbb{Z}l \oplus \mathbb{Z}e$. $\square$

Motivated by the above discussion, we have the following definition:

Definition 2.7 (20.2.8)**.** Let X be a surface over a field k. Let $C \subseteq X$ be a curve consisting of smooth points such that $C \cong \mathbb{P}^1_k$ and its self-intersection number is $(C \cdot C) = -1$. We call such a curve C a (-1)-curve.

A glimpse: X is a smooth projective surface over $\mathbb{C}$. It is called **minimal** if it does not contain any rational (i.e. birational to $\mathbb{P}^1_{\mathbb{C}}$) (-1)-curves. There is a conclusion that if $|mK_X| \neq 0$ for some $m > 0$, then X is minimal if and only if K_X is nef, i.e. $(K_X \cdot C) \geq 0$ for every curve $C \subseteq X$.

Remark. *By analyzing the generators of* $\mathrm{Pic}(\mathbb{P}^1 \times \mathbb{P}^1)$ *and* $\mathrm{Pic}(\mathrm{Bl}_p\mathbb{P}^2)$ *along with their intersection forms, we can deduce that* $\mathbb{P}^1 \times \mathbb{P}^1 \ncong \mathrm{Bl}_p\mathbb{P}^2$.

Classical Example (20.2.G): Note that $l \cong \mathbb{P}^1 \cong e$. Let us take two copies of $\mathrm{Bl}_p\mathbb{P}^2$, denoted as Y_1 and Y_2. In Y_1, we have the naturally embedded curves l_1 and e_1; in Y_2, we have the curves l_2 and e_2. We construct a new space X by gluing Y_1 and Y_2 along these curves such that l_1 is identified with e_2, and e_1 is identified with l_2.

Suppose, for the sake of contradiction, that X is a projective scheme. Then there exists an ample line bundle $\mathcal{L}$ on X. Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be the pullbacks of $\mathcal{L}$ via the natural inclusions $Y_1 \hookrightarrow X$ and $Y_2 \hookrightarrow X$, respectively.

Since the restriction of an ample line bundle to a closed subscheme remains ample, both $\mathcal{L}_1$ and $\mathcal{L}_2$ must still be ample. Consequently, their intersection with any curve in Y_1 and Y_2 must be strictly positive. In particular, for the curves $m_1 \sim l_1 - e_1$ and $m_2 \sim l_2 - e_2$, we have:

$$\mathcal{L}_1 \cdot m_1 > 0 \quad \text{and} \quad \mathcal{L}_2 \cdot m_2 > 0.$$

By linearity, this implies:

$$\mathcal{L}_1 \cdot l_1 > \mathcal{L}_1 \cdot e_1 \quad \text{and} \quad \mathcal{L}_2 \cdot l_2 > \mathcal{L}_2 \cdot e_2.$$

However, due to the gluing conditions, the intersection numbers evaluated on the identified curves must be equal:

$$\mathcal{L}_1 \cdot l_1 = \mathcal{L}_2 \cdot e_2 \quad \text{and} \quad \mathcal{L}_1 \cdot e_1 = \mathcal{L}_2 \cdot l_2.$$

This immediately leads to a contradiction! (We would obtain $\mathcal{L}_1 \cdot l_1 > \mathcal{L}_1 \cdot e_1 = \mathcal{L}_2 \cdot l_2 > \mathcal{L}_2 \cdot e_2 = \mathcal{L}_1 \cdot l_1$, which is impossible).

Therefore, X is not projective. Since X is constructed by gluing proper schemes along proper closed subschemes, X itself is proper. This provides a classical example showing that a proper scheme is not necessarily projective.

3 Hodge Index Theorem

Theorem 3.1. Suppose X is an irreducible smooth projective surface over a field k. Let $\mathcal{L}, \mathcal{H} \in \mathrm{Pic}(X)$ be line bundles such that $\mathcal{H} \cdot \mathcal{H} > 0$ and $\mathcal{L} \cdot \mathcal{H} = 0$. Then:

(a) $\mathcal{L} \cdot \mathcal{L} \leq 0$.

(b) $\mathcal{L} \cdot \mathcal{L} = 0$ if and only if $\mathcal{L} \equiv_{\mathrm{num}} 0$.

***Proof*.** The proof proceeds in several steps.
Step 1:

Lemma 3.2. If $\mathcal{H}$ is an ample line bundle and $\mathcal{M}$ is a line bundle such that $(\mathcal{M} \cdot \mathcal{H}) > (\omega_X \cdot \mathcal{H})$, then $h^2(X, \mathcal{M}) = 0$.

Proof of Lemma 1. By Serre duality, we have $h^2(X, \mathcal{M}) = h^0(X, \omega_X \otimes \mathcal{M}^\vee)$. We prove this by contradiction. Suppose that $h^0(X, \omega_X \otimes \mathcal{M}^\vee) \neq 0$. Then there exists an effective divisor D such that $\mathcal{O}(D) \cong \omega_X \otimes \mathcal{M}^\vee$.

By the ampleness of $\mathcal{H}$ and the effectiveness of D, we must have $(\mathcal{H} \cdot D) > 0$. On the other hand, since $D \equiv \omega_X - \mathcal{M}$, this implies $(\omega_X \cdot \mathcal{H}) > (\mathcal{M} \cdot \mathcal{H})$. This directly contradicts our initial assumption! Thus, $h^2(X, \mathcal{M}) = 0$. $\square$

Step 2:

Lemma 3.3. If $\mathcal{H}$ is an ample line bundle, and $\mathcal{L}$ is a line bundle with $\mathcal{L} \cdot \mathcal{L} > 0$ and $\mathcal{L} \cdot \mathcal{H} > 0$, then for $n \gg 0$, the tensor power $\mathcal{L}^{\otimes n}$ has a non-zero global section (i.e., $h^0(X, \mathcal{L}^{\otimes n}) > 0$).

Proof of Lemma 2. Since $\mathcal{L} \cdot \mathcal{H} > 0$, we can find a sufficiently large integer $n \gg 0$ such that $n(\mathcal{L} \cdot \mathcal{H}) > (\omega_X \cdot \mathcal{H})$. By Lemma 1, this implies $h^2(X, \mathcal{L}^{\otimes n}) = 0$.

Applying the Riemann-Roch theorem for surfaces to $\mathcal{L}^{\otimes n}$, we obtain:

$$\begin{aligned} h^0(X, \mathcal{L}^{\otimes n}) - h^1(X, \mathcal{L}^{\otimes n}) &= \chi(X, \mathcal{L}^{\otimes n}) \\ &= \frac{n^2}{2}(\mathcal{L} \cdot \mathcal{L}) - \frac{n}{2}(\mathcal{L} \cdot \omega_X) + \chi(X, \mathcal{O}_X). \end{aligned}$$

Since $\mathcal{L} \cdot \mathcal{L} > 0$, the quadratic term $\frac{n^2}{2}(\mathcal{L} \cdot \mathcal{L})$ strictly dominates the expression for large n. Therefore, for $n \gg 0$, the right-hand side approaches infinity, which implies $h^0(X, \mathcal{L}^{\otimes n}) - h^1(X, \mathcal{L}^{\otimes n}) > 0$. Since $h^1(X, \mathcal{L}^{\otimes n}) \geq 0$, it immediately follows that $h^0(X, \mathcal{L}^{\otimes n}) > 0$. $\square$

Step 3: We now prove the main theorem for the specific case where $\mathcal{H}$ is ample.

For $n \gg 0$, the line bundle $\mathcal{L} \otimes \mathcal{H}^{\otimes n}$ becomes ample (or very ample). Let us denote this ample line bundle by $\mathcal{H}'$. We proceed to prove the theorem by contradiction. Suppose that $\mathcal{L} \cdot \mathcal{L} > 0$. Compute the intersection:

$$(\mathcal{L} \cdot \mathcal{H}') = (\mathcal{L} \cdot (\mathcal{L} \otimes \mathcal{H}^{\otimes n})) = (\mathcal{L} \cdot \mathcal{L}) + n(\mathcal{L} \cdot \mathcal{H}) = (\mathcal{L} \cdot \mathcal{L}) > 0.$$

By Lemma 2, since $\mathcal{H}'$ is ample, $(\mathcal{L} \cdot \mathcal{L}) > 0$, and $(\mathcal{L} \cdot \mathcal{H}') > 0$, the line bundle $\mathcal{L}^{\otimes m}$ must have a non-zero global section for some integer $m \gg 0$. Let $D \in |\mathcal{L}^{\otimes m}|$ be the corresponding effective divisor. Since $\mathcal{H}$ is ample, its intersection with any effective divisor is strictly positive, hence $(D \cdot \mathcal{H}) > 0$. This implies $\frac{1}{m}(D \cdot \mathcal{H}) = (\mathcal{L} \cdot \mathcal{H}) > 0$. This directly contradicts our initial assumption that $(\mathcal{L} \cdot \mathcal{H}) = 0$! Therefore, we must have $(\mathcal{L} \cdot \mathcal{L}) \leq 0$.

Step 4: The case $(\mathcal{L} \cdot \mathcal{L}) = 0$ when $\mathcal{H}$ is ample. Suppose $(\mathcal{L} \cdot \mathcal{L}) = 0$. We wish to prove that $\mathcal{L} \equiv_{\text{num}} 0$. We proceed by contradiction and suppose $\mathcal{L} \not\equiv_{\text{num}} 0$. By definition, there exists a line bundle $\mathcal{Q}$ such that $(\mathcal{Q} \cdot \mathcal{L}) \neq 0$. Consider the line bundle (written additively):

$$\mathcal{R} := (\mathcal{H} \cdot \mathcal{H})\mathcal{Q} - (\mathcal{Q} \cdot \mathcal{H})\mathcal{H}.$$

We can verify the following intersections:

$$(\mathcal{R} \cdot \mathcal{H}) = (\mathcal{H} \cdot \mathcal{H})(\mathcal{Q} \cdot \mathcal{H}) - (\mathcal{Q} \cdot \mathcal{H})(\mathcal{H} \cdot \mathcal{H}) = 0.$$

$$(\mathcal{R} \cdot \mathcal{L}) = (\mathcal{H} \cdot \mathcal{H})(\mathcal{Q} \cdot \mathcal{L}) - (\mathcal{Q} \cdot \mathcal{H})(\mathcal{H} \cdot \mathcal{L}) = (\mathcal{H} \cdot \mathcal{H})(\mathcal{Q} \cdot \mathcal{L}) \neq 0.$$

Now, define $\mathcal{L}' := \mathcal{R} \otimes \mathcal{L}^{\otimes m}$ (additively $\mathcal{R} + m\mathcal{L}$). We have:

$$(\mathcal{L}' \cdot \mathcal{H}) = (\mathcal{R} \cdot \mathcal{H}) + m(\mathcal{L} \cdot \mathcal{H}) = 0 + 0 = 0.$$

By the conclusion of Step 3, since $(\mathcal{L}' \cdot \mathcal{H}) = 0$, it must follow that $(\mathcal{L}' \cdot \mathcal{L}') \leq 0$. However, let us compute $(\mathcal{L}' \cdot \mathcal{L}')$ directly:

$$(\mathcal{L}' \cdot \mathcal{L}') = (\mathcal{R} \cdot \mathcal{R}) + 2m(\mathcal{R} \cdot \mathcal{L}) + m^2(\mathcal{L} \cdot \mathcal{L}) = (\mathcal{R} \cdot \mathcal{R}) + 2m(\mathcal{R} \cdot \mathcal{L}).$$

Because $(\mathcal{R} \cdot \mathcal{L}) \neq 0$, the term $2m(\mathcal{R} \cdot \mathcal{L})$ is a non-constant linear function of m. We can choose some integer $m \in \mathbb{Z}$ such that $(\mathcal{R} \cdot \mathcal{R}) + 2m(\mathcal{R} \cdot \mathcal{L}) > 0$. This yields $(\mathcal{L}' \cdot \mathcal{L}') > 0$, which contradicts the conclusion $(\mathcal{L}' \cdot \mathcal{L}') \leq 0$. Therefore, $\mathcal{L} \equiv_{\text{num}} 0$. This completes the proof of the Hodge Index Theorem for the case where $\mathcal{H}$ is ample.

Step 5: The general case. Now suppose $\mathcal{H}$ is not necessarily ample, but only satisfies

$(\mathcal{H} \cdot \mathcal{H}) > 0$. Let $\mathcal{A}$ be an arbitrary ample line bundle on X. If $(\mathcal{A} \cdot \mathcal{L}) = 0$, then by applying the established ample case to $\mathcal{A}$, we immediately get $(\mathcal{L} \cdot \mathcal{L}) \leq 0$ (and $(\mathcal{L} \cdot \mathcal{L}) = 0$ if and only if $\mathcal{L} \equiv_{\text{num}} 0$). The theorem holds. If $(\mathcal{A} \cdot \mathcal{L}) \neq 0$, then trivially $\mathcal{L} \not\equiv_{\text{num}} 0$. Also, note that if $(\mathcal{A} \cdot \mathcal{H}) = 0$, applying the ample case to $\mathcal{A}$ would yield $(\mathcal{H} \cdot \mathcal{H}) \leq 0$, which contradicts the given condition $(\mathcal{H} \cdot \mathcal{H}) > 0$. Thus, we must have $(\mathcal{A} \cdot \mathcal{H}) \neq 0$. Consider the line bundle:

$$\mathcal{M} := (\mathcal{A} \cdot \mathcal{L})\mathcal{H} - (\mathcal{A} \cdot \mathcal{H})\mathcal{L}.$$

Intersecting with $\mathcal{A}$, we get:

$$(\mathcal{A} \cdot \mathcal{M}) = (\mathcal{A} \cdot \mathcal{L})(\mathcal{A} \cdot \mathcal{H}) - (\mathcal{A} \cdot \mathcal{H})(\mathcal{A} \cdot \mathcal{L}) = 0.$$

Since $\mathcal{A}$ is ample and $(\mathcal{A} \cdot \mathcal{M}) = 0$, we can apply the ample case to deduce that $(\mathcal{M} \cdot \mathcal{M}) \leq 0$. Let us expand $(\mathcal{M} \cdot \mathcal{M})$ using the bilinearity of the intersection form:

$$(\mathcal{M} \cdot \mathcal{M}) = (\mathcal{A} \cdot \mathcal{L})^2(\mathcal{H} \cdot \mathcal{H}) - 2(\mathcal{A} \cdot \mathcal{L})(\mathcal{A} \cdot \mathcal{H})(\mathcal{H} \cdot \mathcal{L}) + (\mathcal{A} \cdot \mathcal{H})^2(\mathcal{L} \cdot \mathcal{L}) \leq 0.$$

Since $(\mathcal{L} \cdot \mathcal{H}) = 0$ by assumption, the middle cross-term vanishes, leaving:

$$\underbrace{(\mathcal{A} \cdot \mathcal{L})^2}_{>0}\underbrace{(\mathcal{H} \cdot \mathcal{H})}_{>0} + \underbrace{(\mathcal{A} \cdot \mathcal{H})^2}_{>0}(\mathcal{L} \cdot \mathcal{L}) \leq 0.$$

The first term is strictly positive because $(\mathcal{A} \cdot \mathcal{L}) \neq 0$ and $(\mathcal{H} \cdot \mathcal{H}) > 0$. The coefficient $(\mathcal{A} \cdot \mathcal{H})^2$ is also strictly positive. For the entire sum to be non-positive, we must necessarily have:

$$(\mathcal{L} \cdot \mathcal{L}) < 0.$$

This concludes the proof of the Hodge Index Theorem for the general case.

□

We introduce the following interesting consequence:

Lemma 3.4 (Neron − Severi group)**.** $N_1(X) := \mathrm{Div}(X)/\sim$, where $\sim$ is the numerical equivalence relation. Then $N_1(X)$ is a free abelian group of finite rank.

Corollary 3.5. Let X be an irreducible smooth projective surface over a field k. Let ρ be the Picard number of X.

(a) In the rational Néron-Severi space $N^1(X)_{\mathbb{Q}} = (\mathrm{Pic}(X)/\equiv_{\text{num}}) \otimes \mathbb{Q}$, there exists an orthogonal basis $e_1, e_2, \ldots, e_\rho$ such that the intersection form is diagonal with exactly one strictly positive entry and $\rho - 1$ strictly negative entries.

(b) In the real Néron-Severi space $N^1(X)_{\mathbb{R}} = N^1(X)_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R}$, there exists an orthogonal basis $u_1, u_2, \ldots, u_\rho$ such that the intersection matrix is exactly $\operatorname{diag}(1, -1, \ldots, -1)$.

***Proof*.** **Proof of (a): The $N^1(X)_{\mathbb{Q}}$ case.**

Since X is a projective surface, there exists an ample line bundle $\mathcal{H}$. Let $e_1 = [\mathcal{H}] \in N^1(X)_{\mathbb{Q}}$ be its numerical class. Because $\mathcal{H}$ is ample, its self-intersection is strictly positive, meaning $e_1 \cdot e_1 > 0$.

We define the orthogonal complement of e_1 in $N^1(X)_{\mathbb{Q}}$ with respect to the intersection form:

$$e_1^{\perp} = \{D \in N^1(X)_{\mathbb{Q}} \mid D \cdot e_1 = 0\}.$$

Because $e_1 \cdot e_1 \neq 0$, the intersection form is non-degenerate along the one-dimensional subspace spanned by e_1. Therefore, we have an orthogonal direct sum decomposition of the ρ-dimensional vector space $N^1(X)_{\mathbb{Q}}$:

$$N^1(X)_{\mathbb{Q}} = \mathbb{Q}e_1 \oplus e_1^{\perp}.$$

The dimension of $e_1^{\perp}$ is exactly $\rho - 1$.

By the Hodge Index Theorem, for any non-zero element $L \in e_1^{\perp}$, we have $L \cdot e_1 = 0$ and $L \not\equiv_{\text{num}} 0$. The theorem guarantees that $L \cdot L < 0$. This implies that the restriction of the intersection form to the subspace $e_1^{\perp}$ is strictly negative definite.

Since every definite symmetric bilinear form over a field of characteristic zero admits an orthogonal basis (via the Gram-Schmidt orthogonalization process), there exists an orthogonal basis $e_2, \ldots, e_\rho$ for $e_1^{\perp}$.

Therefore, $\{e_1, e_2, \ldots, e_\rho\}$ forms an orthogonal basis for the entire space $N^1(X)_{\mathbb{Q}}$. Under this basis, the intersection matrix is diagonal. The diagonal entries are $e_1 \cdot e_1 > 0$ and $e_i \cdot e_i < 0$ for $i = 2, \ldots, \rho$. Thus, the signature of the intersection form is $(+, -, \ldots, -)$.

Proof of (b): The $N^1(X)_{\mathbb{R}}$ case.

We now extend the scalars to the real numbers, considering the space $N^1(X)_{\mathbb{R}}$. The orthogonal basis $\{e_1, e_2, \ldots, e_\rho\}$ found in part (a) naturally serves as an orthogonal basis for $N^1(X)_{\mathbb{R}}$.

Over the real numbers, we can normalize these basis vectors because every positive real number has a well-defined real square root. We define a new basis $\{u_1, u_2, \ldots, u_\rho\}$ by scaling the previous basis vectors:

$$u_1 = \frac{1}{\sqrt{e_1 \cdot e_1}} e_1,$$

and for $i = 2, \ldots, \rho$,

$$u_i = \frac{1}{\sqrt{-(e_i \cdot e_i)}} e_i.$$

We can compute the self-intersections of this new basis. For u_1, we have:

$$u_1 \cdot u_1 = \left(\frac{1}{\sqrt{e_1 \cdot e_1}}\right)^2 (e_1 \cdot e_1) = \frac{e_1 \cdot e_1}{e_1 \cdot e_1} = 1.$$

For $i \geq 2$, since $e_i \cdot e_i < 0$, the quantity $-(e_i \cdot e_i)$ is strictly positive, so its real square root is well-defined. We have:

$$u_i \cdot u_i = \left(\frac{1}{\sqrt{-(e_i \cdot e_i)}}\right)^2 (e_i \cdot e_i) = \frac{e_i \cdot e_i}{-(e_i \cdot e_i)} = -1.$$

Since scaling vectors does not affect their mutual orthogonality, $\{u_1, \ldots, u_\rho\}$ is an orthogonal basis for $N^1(X)_{\mathbb{R}}$. Under this normalized basis, the intersection matrix is exactly the diagonal matrix $\mathrm{diag}(1, -1, \ldots, -1)$. $\square$

4 The Hirzebruch Surfaces

Lemma 4.1 (20.2.I)**.** X is a scheme, $\mathcal{V}$ is a rank 2 locally free sheaf on X. $\mathbb{P}\mathcal{V} := \mathbf{Proj}_{\mathcal{O}_X}(\mathrm{Sym}^\bullet \mathcal{V}^\vee)$. $\sigma : X \to \mathbb{P}\mathcal{V}$ is a section of the projection $\pi : \mathbb{P}\mathcal{V} \to X$. Then the normal bundle $\mathcal{N}_{\sigma(X)/\mathbb{P}\mathcal{V}} := \sigma^*\mathcal{O}_{\mathbb{P}\mathcal{V}}(\sigma(X))$ is isomorphic to $\mathcal{L}^{\otimes 2} \otimes \det \mathcal{V}$, where $\mathcal{L} := \sigma^*\mathcal{O}_{\mathbb{P}\mathcal{V}}(1)$.

Remark. *In the original description in [FOAG] 20.2.I, by* ***17.2.I****, one section always corresponds to an exact sequence* $0 \to \mathscr{S} \to \mathscr{V} \to \mathscr{Q} \to 0$. $\mathscr{S}$ *corresponds to* $\mathcal{L}^\vee$, $\mathscr{S}^\vee \otimes \mathscr{Q} = \mathscr{S}^{\vee\otimes 2} \otimes (\mathscr{S} \otimes \mathscr{Q}) = \mathscr{S}^{\vee\otimes 2} \otimes \det \mathscr{V}$ *corresponds to* $\mathcal{L}^{\otimes 2} \otimes \det \mathcal{V}$ *here. We will not follow the idea in [FOAG]. If you first read it, you can skip and acknowledge it.*

Proof. **1. Setup and Trivialization**
Choose an affine open cover $X = \bigcup U_i$ where $U_i = \mathrm{Spec}\, A_i$, such that the cover is **subordinate to both $\mathcal{V}^\vee$ and $\mathcal{L}$.**

- Over U_i, $\mathcal{V}^\vee$ is trivialized with basis $(s_{i0}^\vee, s_{i1}^\vee)$. On the overlap $U_i \cap U_j$, the transition matrix is $M_{ij} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$. Thus:

$$\begin{aligned} s_{i0}^\vee &= A s_{j0}^\vee + C s_{j1}^\vee \\ s_{i1}^\vee &= B s_{j0}^\vee + D s_{j1}^\vee \end{aligned}$$

  The transition function for $\det \mathcal{V}^\vee$ is $\det(M_{ij}) = AD - BC$.

- Over U_i, $\mathcal{L}$ is trivialized with basis e_i. On the overlap, $e_i = c_{ij} e_j$, meaning the transition function for $\mathcal{L}$ is c_{ij}.

2. The Section and its Compatibility
Locally over U_i, this acts on the basis as:

$$s_{i0}^\vee \mapsto b_{i0} e_i, \quad s_{i1}^\vee \mapsto b_{i1} e_i$$

To be compatible with the transition functions on $U_i \cap U_j$, substituting $e_i = c_{ij} e_j$ yields the relations for the coefficients:

$$\begin{aligned} c_{ij} b_{i0} &= A b_{j0} + C b_{j1} \\ c_{ij} b_{i1} &= B b_{j0} + D b_{j1} \end{aligned}$$

Because $\mathcal{O}_{\mathbb{P}\mathcal{V}}(1)|_{D_+(s_{i0}^\vee)} = \widetilde{A_i\left[\frac{s_{i1}^\vee}{s_{i0}^\vee}\right]} \cdot s_{i0}^\vee$, $\sigma^* s_{i0}^\vee$ is also a generator of $\mathcal{L}|_{\sigma^{-1}(D_+(s_{i0}^\vee))}$, hence b_{i0} is a unit in $\Gamma(\sigma^{-1}(D_+(s_{i0}^\vee)), \mathcal{O}_{U_i})$. Similarly, b_{i1} is a unit in $\Gamma(\sigma^{-1}(D_+(s_{i1}^\vee)), \mathcal{O}_{U_i})$.

3. The Homogeneous Ideal of the Section

In the projective bundle $\mathbb{P}\mathcal{V}$, the section $\sigma(X)$ is defined locally over U_i by the kernel of the surjection, which is generated by the homogeneous linear form:

$$F_i = b_{i1} s_{i0}^\vee - b_{i0} s_{i1}^\vee$$

Let us compute the relation between F_i and F_j on overlaps. Expanding $c_{ij} F_i$:

$$c_{ij} F_i = (c_{ij} b_{i1}) s_{i0}^\vee - (c_{ij} b_{i0}) s_{i1}^\vee$$

Substitute the transition relations for $s^\vee$ and b:

$$c_{ij} F_i = (B b_{j0} + D b_{j1})(A s_{j0}^\vee + C s_{j1}^\vee) - (A b_{j0} + C b_{j1})(B s_{j0}^\vee + D s_{j1}^\vee)$$

Expanding and canceling the AB and CD terms leaves:

$$c_{ij} F_i = (AD - BC)(b_{j1} s_{j0}^\vee - b_{j0} s_{j1}^\vee) = \det(M_{ij}) F_j$$

Thus, the transition relation for the homogeneous generator is:

$$F_i = c_{ij}^{-1} \det(M_{ij}) F_j$$

4. Local Affine Generators of the Conormal Bundle

To find the conormal bundle $\mathcal{I}/\mathcal{I}^2$, we must pass to affine charts.

- In the standard affine chart $D_+(s_{i0}^\vee) \cong \operatorname{Spec} A_i\left[\frac{s_{i1}^\vee}{s_{i0}^\vee}\right]$, we divide F_i by the invertible element $s_{i0}^\vee$ to get the local generator:

$$\Upsilon_{i0} = \frac{F_i}{s_{i0}^\vee} = b_{i1} - b_{i0} \frac{s_{i1}^\vee}{s_{i0}^\vee}$$

- In the other affine chart $D_+(s_{i1}^\vee) \cong \operatorname{Spec} A_i\left[\frac{s_{i0}^\vee}{s_{i1}^\vee}\right]$, we divide by $s_{i1}^\vee$:

$$\Upsilon_{i1} = \frac{F_i}{s_{i1}^\vee} = b_{i1} \frac{s_{i0}^\vee}{s_{i1}^\vee} - b_{i0}$$

On their overlap $D_+(s_{i0}^\vee s_{i1}^\vee)$, they are related by:

$$\Upsilon_{i0} = \frac{s_{i1}^\vee}{s_{i0}^\vee}\Upsilon_{i1}$$

5. Gluing the Global Generator over $\operatorname{Spec} A_i$
We pull these generators back to the base space X via σ^*. Since $\sigma^*(s_{i0}^\vee) = b_{i0}$ and $\sigma^*(s_{i1}^\vee) = b_{i1}$, the relation becomes:

$$\sigma^*\Upsilon_{i0} = \frac{b_{i1}}{b_{i0}}\sigma^*\Upsilon_{i1}$$

Cross-multiplying eliminates the fractions:

$$b_{i0}\sigma^*\Upsilon_{i0} = b_{i1}\sigma^*\Upsilon_{i1}$$

Let us define this common element as Φ_i. Because b_{i0} (resp. b_{i1}) is a unit in $\Gamma(\sigma^{-1}(D_+(s_{i0}^\vee)), \mathcal{O}_{U_i})$ (resp. $\Gamma(\sigma^{-1}(D_+(s_{i1}^\vee)), \mathcal{O}_{U_i})$), $b_{i0}\sigma^*\Upsilon_{i0}$ and $b_{i1}\sigma^*\Upsilon_{i1}$ are both generators of the conormal bundle over their respective affine charts. Since they agree on the intersection, Φ_i **is a nowhere-vanishing global generator for the conormal bundle over the entirety of** $\operatorname{Spec} A_i$.

6. Calculating the Transition Function for the Conormal Bundle
Now we determine how Φ_i transforms to Φ_j on $U_i \cap U_j$. We first relate the local affine generators Υ_{i0} and Υ_{j0}:

$$\begin{aligned}\Upsilon_{i0} = \frac{F_i}{s_{i0}^\vee} &= \frac{c_{ij}^{-1}\det(M_{ij})F_j}{s_{i0}^\vee} \\ &= c_{ij}^{-1}\det(M_{ij})\frac{s_{j0}^\vee}{s_{i0}^\vee}\left(\frac{F_j}{s_{j0}^\vee}\right) \\ &= c_{ij}^{-1}\det(M_{ij})\frac{s_{j0}^\vee}{s_{i0}^\vee}\Upsilon_{j0}\end{aligned}$$

Pulling back to the section $\sigma(X)$, and noting that $\sigma^*\left(\frac{s_{j0}^\vee}{s_{i0}^\vee}\right) = \frac{b_{j0}}{c_{ij}b_{i0}}$, we get:

$$\sigma^*\Upsilon_{i0} = c_{ij}^{-1}\det(M_{ij})\left(\frac{b_{j0}}{c_{ij}b_{i0}}\right)\sigma^*\Upsilon_{j0} = c_{ij}^{-2}\det(M_{ij})\frac{b_{j0}}{b_{i0}}\sigma^*\Upsilon_{j0}$$

To recover our glued global generators Φ, we multiply both sides by b_{i0}:

$$b_{i0}\sigma^*\Upsilon_{i0} = c_{ij}^{-2}\det(M_{ij})b_{j0}\sigma^*\Upsilon_{j0}$$

Substituting the definition of Φ, the local variables b_{i0} and b_{j0} are perfectly absorbed:

$$\Phi_i = c_{ij}^{-2} \det(M_{ij})\Phi_j$$

7. Conclusion
The element Φ_i generates the conormal bundle $\mathcal{O}(-\sigma(X))|_X$ (the ideal sheaf modulo its square) over U_i. The transition function g_{ij}^{conormal} is therefore $c_{ij}^{-2} \det(M_{ij})$.

- c_{ij}^{-2} is the transition function for $\mathcal{L}^{\otimes -2}$.
- $\det(M_{ij})$ is the transition function for $\det \mathcal{V}^\vee$.

Thus, $\mathcal{O}(-\sigma(X))|_X \cong \mathcal{L}^{\otimes -2} \otimes \det \mathcal{V}^\vee$. Therefore, $\mathcal{N}_{\sigma(X)/\mathbb{P}\mathcal{V}} \cong \mathcal{L}^{\otimes 2} \otimes \det \mathcal{V}$. □

Definition 4.2 (Hirzebruch surface)**.** We define the **Hirzebruch surface** $\mathbb{F}_n$ as the projective bundle

$$\mathbb{F}_n := \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(n))$$

equipped with the natural projection $\pi : \mathbb{F}_n \to \mathbb{P}^1$.

Consider a section $\sigma : \mathbb{P}^1 \to \mathbb{F}_n$ such that $\sigma^*\mathcal{O}_{\mathbb{F}_n}(1) = \mathcal{O}_{\mathbb{P}^1}(-n)$, which yields the following natural short exact sequence:

$$0 \to \mathcal{O}_{\mathbb{P}^1}(n) \to \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(n) \to \mathcal{O}_{\mathbb{P}^1} \to 0$$

(The existence follows from **17.2.I**.)

Let $\mathcal{N}$ denote the normal bundle of this section. By applying the previously established isomorphism for normal bundles of sections in projective bundles, we find that:

$$\mathcal{N} \cong \mathcal{O}_{\mathbb{P}^1}(-n)$$

We define the curve $E \subseteq \mathbb{F}_n$ to be the image of this section, namely $E := \sigma(\mathbb{P}^1)$. The self-intersection number of the curve E on the surface $\mathbb{F}_n$ is given by the degree of its normal bundle. Therefore, we obtain:

$$(E \cdot E) = -n$$

Let us consider another section $\sigma' : \mathbb{P}^1 \to \mathbb{F}_n$ such that $\sigma'^*\mathcal{O}_{\mathbb{F}_n}(1) = \mathcal{O}_{\mathbb{P}^1}$. This corresponds to the following exact sequence:

$$0 \to \mathcal{O}_{\mathbb{P}^1} \to \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(n) \to \mathcal{O}_{\mathbb{P}^1}(n) \to 0$$

Let C denote the image of this section. Its self-intersection number is given by:

$$(C \cdot C) = \deg(\mathcal{O}_{\mathbb{P}^1}^{\vee} \otimes \mathcal{O}_{\mathbb{P}^1}(n)) = n$$

Let p be any k-point of $\mathbb{P}^1$, and let $F = \pi^{-1}(p)$ denote the fiber over p.

There is an **observation (20.2.K)** that: The line bundle $\mathcal{O}(F)$ is independent of the choice of the point p.

Proof of the observation. Since the projection morphism π is flat, the pullback of the Weil divisor $\pi^*\{p\}$ is well-defined[2]. We have $\mathcal{O}(F) = \mathcal{O}(\pi^*\{p\}) \cong \pi^*\mathcal{O}_{\mathbb{P}^1}(\{p\}) \cong \pi^*\mathcal{O}_{\mathbb{P}^1}(1)$. This shows that $\mathcal{O}(F)$ depends only on the degree of the point, making it independent of the specific choice of p. □

Proposition 4.3 (20.2.L)**.** The Picard group $\mathrm{Pic}(\mathbb{F}_n)$ is generated by the classes of E and F.

Proof Sketch. We outline the idea of the proof. The base $\mathbb{P}^1$ can be covered by two affine open patches $\operatorname{Spec} k[x]$ and $\operatorname{Spec} k[y]$, glued together via $x \leftrightarrow y^{-1}$. Over $\operatorname{Spec} k[x]$, the projective bundle is locally given by $\operatorname{Proj} k[x][Z_0, Z_1]$, where Z_0 corresponds to the dual of $1 \in \Gamma(\operatorname{Spec} k[x], \mathcal{O}_{\mathbb{P}^1})$ and Z_1 corresponds to the dual of a local generator in $\Gamma(\operatorname{Spec} k[x], \mathcal{O}_{\mathbb{P}^1}(n))$.

We can restrict our attention to an affine open subscheme $\operatorname{Spec} k[x, Z_1/Z_0] \subseteq \operatorname{Proj} k[x][Z_0, Z_1] \subseteq \mathbb{F}_n$. The closed subset defined by $V_+(Z_0)$ corresponds to the subbundle $\mathcal{O}_{\mathbb{P}^1}(n) \hookrightarrow \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(n)$, which is exactly the negative section E.

Note that the complement of this affine chart in $\mathbb{F}_n$, namely $\mathbb{F}_n \setminus \operatorname{Spec} k[x, Z_1/Z_0]$, consists of $E = V_+(Z_0)$ union the preimage under π of the point at infinity of $\mathbb{P}^1$ (which corresponds to the origin in $\operatorname{Spec} k[y]$). This preimage is simply a fiber F.

Since the Picard group of the affine space $\operatorname{Spec} k[x, Z_0/Z_1]$ is trivial, the excision exact sequence implies that $\mathrm{Pic}(\mathbb{F}_n)$ is generated by the classes of the irreducible components of the complement, which are E and F. By a similar argument, the Picard group is also generated by C and F. □

Now let's **calculate (20.2.M)** the intersection numbers for these classes.

By choosing two distinct points $p, q \in \mathbb{P}^1$ (for example, the origin of $\operatorname{Spec} k[x]$ and the origin of $\operatorname{Spec} k[y]$), their corresponding fibers F_p and F_q are disjoint, meaning $F_p \cap F_q = \emptyset$. Therefore, the self-intersection of the fiber class is zero:

$$(F \cdot F) = (F_p \cdot F_q) = 0$$

[2]You can refer to Liu Qing's textbook.

From previous results, we already know that the self-intersection of the negative section is:

$$(E \cdot E) = -n$$

Finally, since the embedding $E \hookrightarrow \mathbb{F}_n$ acts as a section of the projection π, it intersects every fiber in exactly one point. Therefore, the intersection number is:

$$(E \cdot F) = \#(E \cap F) = 1$$

This intersection corresponds to a single point on $E \cong \mathbb{P}^1$.

Since the Picard group is generated by E and F, by linear independence gained by the computation above, we have an isomorphism $\operatorname{Pic}(\mathbb{F}_n) \cong \mathbb{Z}[E] \oplus \mathbb{Z}[F]$. Thus, we can write the class of the curve C as:

$$[C] = a[E] + b[F]$$

Using the intersection numbers, we can solve for a and b:

$$(C \cdot F) = 1 \implies a(E \cdot F) + b(F \cdot F) = a = 1$$
$$(C \cdot C) = n \implies a^2(E \cdot E) + 2ab(E \cdot F) + b^2(F \cdot F) = 1(-n) + 2b(1) = n \implies b = n$$

Therefore, the class of C is $[C] = [E] + n[F]$.

Corollary 4.4 (20.2.O)**.** The dimension of the space of global sections satisfies $h^0(\mathbb{F}_n, \mathcal{O}_{\mathbb{F}_n}(C)) > 1$.

Proof. This follows because C is an effective curve with linear equivalence class $C \sim E + nF$. □

Proposition 4.5 (20.2.P)**.** Every effective curve on $\mathbb{F}_n$ is a non-negative linear combination of E and F. That is, for every effective curve D, there exist non-negative integers $a, b \in \mathbb{Z}_{\geq 0}$ such that $D \sim aE + bF$.

Proof. It suffices to show the case when D is irreducible.

If there exists a closed point p such that $D = \pi^{-1}(p)$, then $D \sim F$ and we are done.

Otherwise, since $\operatorname{Pic}(\mathbb{F}_n) \cong \mathbb{Z}[E] \oplus \mathbb{Z}[F]$, we can write the class of D as $D \sim aE + bF$. Because D is not a fiber, its intersection with a general fiber F is non-negative:

$$(D \cdot F) \geq 0 \implies a(E \cdot F) + b(F \cdot F) = a \geq 0$$

By Corollary 20.2.O, we can find another effective curve $C' \sim C$ different from D, hence such that $(D \cdot C') \geq 0$. Since C' is linearly equivalent to C, we have $(D \cdot C) \geq 0$. Calculating this intersection yields:

$$(D \cdot C) = (aE + bF) \cdot (E + nF) = a(-n) + an + b = b$$

Thus, we conclude that $b \geq 0$. □

Corollary 4.6 (20.2.Q)**.** The Hirzebruch surfaces $\mathbb{F}_n$ are pairwise non-isomorphic.

Proof. Suppose there exists an isomorphism $\psi : \mathbb{F}_n \xrightarrow{\cong} \mathbb{F}_m$.

We know the structure of their respective Picard groups and intersection pairings:

- $\mathrm{Pic}(\mathbb{F}_n) = \mathbb{Z}[E_n] \oplus \mathbb{Z}[F_n]$ with $(E_n \cdot E_n) = -n$, $(E_n \cdot F_n) = 1$, and $(F_n \cdot F_n) = 0$.
- $\mathrm{Pic}(\mathbb{F}_m) = \mathbb{Z}[E_m] \oplus \mathbb{Z}[F_m]$ with $(E_m \cdot E_m) = -m$, $(E_m \cdot F_m) = 1$, and $(F_m \cdot F_m) = 0$.

The isomorphism ψ induces an invertible linear map on the Picard groups:

$$\begin{bmatrix} \psi^* E_m \\ \psi^* F_m \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} E_n \\ F_n \end{bmatrix}$$

and its inverse map gives:

$$\begin{bmatrix} (\psi^{-1})^* E_n \\ (\psi^{-1})^* F_n \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} E_m \\ F_m \end{bmatrix}$$

where the determinant $ad - bc = 1$.

By Proposition 20.2.P, isomorphisms map effective curves to effective curves, so all coefficients in both transformation matrices must be non-negative. Therefore, we must have $a, b, c, d \geq 0$ as well as $-b \geq 0$ and $-c \geq 0$.

This implies $b = c = 0$. Since $ad - bc = 1$ and $a, d \geq 0$, it follows that $a = d = 1$.

Thus, $\psi^* E_m = E_n$. Since isomorphisms preserve intersection numbers, we have:

$$(E_m \cdot E_m) = (\psi^* E_m \cdot \psi^* E_m) = (E_n \cdot E_n)$$

Substituting the self-intersection values yields $-m = -n$, which forces $m = n$. □

Proposition 4.7 (20.2.S)**.** The nef cone of the Hirzebruch surface $\mathbb{F}_n$, denoted $\mathrm{Nef}(\mathbb{F}_n)$, is generated by the classes C and F.

Proof. By the definition of a nef divisor, a divisor class D is nef if and only if its intersection number with every irreducible curve I is non-negative, that is, $(D \cdot I) \geq 0$.

By Proposition 20.2.P, every effective curve (and thus every irreducible curve) I can be expressed as a non-negative linear combination of E and F.

Therefore, by linearity of the intersection product, to check if a divisor D is nef, it suffices to check its intersection with the base generators E and F. Thus, D is nef if and only if:

$$(D \cdot E) \geq 0 \quad \text{and} \quad (D \cdot F) \geq 0$$

Since the Picard group $\mathrm{Pic}(\mathbb{F}_n)$ is generated by E and F, and we know that $C \sim E + nF$, the classes C and F also form a basis. We can therefore write any divisor class D as a linear combination of C and F:

$$D \sim aC + bF$$

which is linearly equivalent to $aE + (b + an)F$.

Let us compute the intersection numbers of D with F and E:

$$(D \cdot F) = (aC + bF) \cdot F = a(C \cdot F) + b(F \cdot F) = a(1) + 0 = a$$
$$(D \cdot E) = (aC + bF) \cdot E = a(C \cdot E) + b(F \cdot E) = a(0) + b(1) = b$$

(Note that we used the relation $(C \cdot E) = (E + nF) \cdot E = (E \cdot E) + n(F \cdot E) = -n + n(1) = 0$).

Consequently, the nef conditions $(D \cdot F) \geq 0$ and $(D \cdot E) \geq 0$ are exactly equivalent to:

$$a \geq 0 \quad \text{and} \quad b \geq 0$$

This proves that a divisor D is nef if and only if it is a non-negative linear combination of C and F. Hence, the nef cone is generated by C and F. $\square$